📊 HR Analytics & Employee Attrition Report

HR Analytics & Employee Attrition Intelligence Overview

1. 🎯 Project Scope & Overview

This executive report delivers an in-depth analysis of employee turnover patterns across 1,470 workforce records. The study evaluates key organizational drivers behind employee attrition, identifies high-risk workforce demographics, and provides actionable HR strategies to enhance employee retention and lower replacement costs.

2. ❓ Key Analytical Questions & Detailed Responses

Question 1: What is the overall organizational attrition rate, and which age groups exhibit the highest turnover?

Detailed Response: The overall organizational attrition rate stands at 16.1%. Demographic profiling indicates that younger professionals aged 26–35 record the highest turnover rate, largely driven by career progression pursuits and competitive entry-level market opportunities.

Strategic Recommendation: Establish clear internal career progression pathways and mentorship programs for early-to-mid career staff during their initial 3 years.

Question 2: How does monthly income and salary bracket correlate with employee resignation decisions?

Detailed Response: Turnover is heavily concentrated among lower income brackets (< $5,000/month). Attrition rates decrease significantly as monthly income rises, confirming financial compensation as a pivotal retention driver for entry-level roles.

Strategic Recommendation: Conduct market-salary benchmarking and implement performance-driven bonus structures to improve compensation competitiveness.

Question 3: Which specific job roles and departments display the highest vulnerability to attrition?

Detailed Response: Operational roles such as Laboratory Technicians and Sales Representatives exhibit the highest turnover rates, whereas managerial and executive roles reflect high stability.

Strategic Recommendation: Enhance work-life balance and incentive programs specifically tailored for high-strain operational and sales roles.

Question 4: What is the impact of overtime work on employee burnout and turnover?

Detailed Response: Employees required to work frequent overtime display double the attrition rate compared to non-overtime staff, signaling potential workplace burnout issues.

Strategic Recommendation: Cap mandatory overtime hours and establish employee wellness initiatives to improve overall retention.

3. 🛠️ Methodology & Analytical Workflow

Phase	Tasks Completed & Technical Explanation
Data Cleansing	Cleaned workforce data, transformed missing entries, and categorized age & salary groups in Power Query.
DAX Modeling	Built DAX measures for Attrition Rate %, Average Monthly Income, Tenure Length, and Departmental Headcount.
Interactive Visualization	Created a Power BI HR Dashboard featuring dynamic slicers for department, age group, and job satisfaction tiers.